Android A/B System Update Guide

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Preface

Overview

The document mainly describes Rockchip A/B System instruction.

Product Version

Chip Name	Kernel Version
RK3326/RK3399/RK356x/RK3588	4.19, 5.10

Intended Audience

This document (this guide) is mainly intended for

Technical Support Engineer

Software Development Engineer

Revision History

Version	Author	Date	Change Description
V1.0.0	Ji Dayao	2022-02-25	Initial version release
V1.1.0	Ji Dayao	2023-02-06	Added configuration description for Android 13
V1.2.0	Ji Dayao	2023-10-29	Added configuration description for Android 14
V1.3.0	Ji Dayao	2024-12-12	Updated Sections 3 and 6.2

Contents

Android A/B System Update Guide

1. Overview
2. System Configuration Instruction
    - 2.1 Configuration Guide for Android System
        - 2.1.1 Android 12 System Configuration
        - 2.1.2 Android 13 System Configuration
        - 2.1.3 Android >=14 System Configuration
    - 2.2 Kernel DTS Configuration
    - 2.3 U-Boot Configuration
    - 2.4 Add Partitions via Advanced Flashing Tool
3. AB System and OTA Package Compilation
4. Validation Methods (Client and Server)
5. Notes
6. A/B miniloader OTA
    - 6.1 rkloader Upgrade Package Build
    - 6.2 Upgrade Command Reference

1. Overview

This document describes the instruction for the Rockchip A/B System, which can be used on Rockchip Android >= 12 platforms.

Using this upgrade method, all partitions to be upgraded (excluding the "super" partition) have two copies (A and B). Compared with the traditional Recovery-based upgrade method, this approach significantly increases storage space requirements. The advantage of A/B upgrades is that they provide seamless upgrades, which can be performed while the Android system is running, based on the interaction strategies and protocols between the customized client and server to complete the upgrade process.

2. System Configuration Instruction

The Android A/B system feature is disabled by default on the Rockchip Android platform. Enabling it requires configuration across three aspects: the Android system, U-BOOT, and kernel DTS. The following provides a detailed instruction.

2.1 Configuration Guide for Android System

2.1.1 Android 12 System Configuration

Configuration of Android 12 system includes:

1. In device\rockchip\rkxxx\BoardConfig.mk enable the AB update configuration item

Set the BOARD_USES_AB_IMAGE configuration to true, defaults to false. Taking rk3326 as an example:

```
vim device/rockchip/rk3326_s/BoardConfig.mk
#AB image definition
-BOARD_USES_AB_IMAGE := false
+BOARD_USES_AB_IMAGE := true
```

Note: If you want to enable the virtual AB feature (not recommended), both BOARD_USES_AB_IMAGE and BOARD_ROCKCHIP_VIRTUAL_AB_ENABLE need to be configured as true, i.e.:

```
-BOARD_USES_AB_IMAGE := false
-BOARD_ROCKCHIP_VIRTUAL_AB_ENABLE := false
+BOARD_USES_AB_IMAGE := true
+BOARD_ROCKCHIP_VIRTUAL_AB_ENABLE := true
```

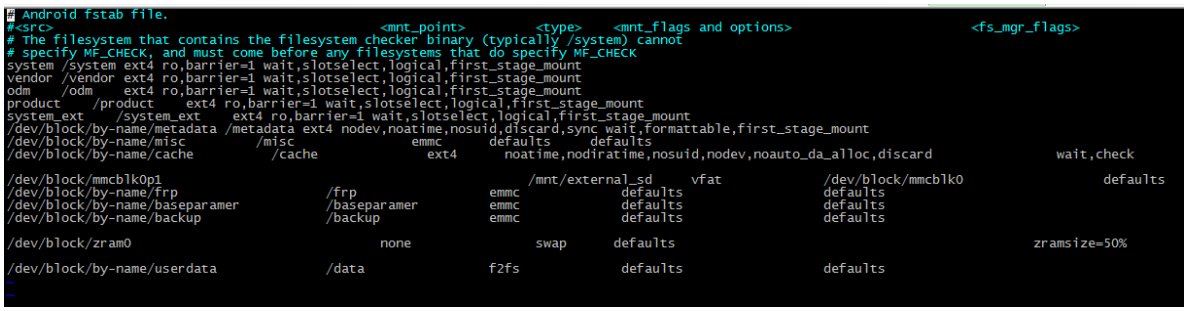
2. In the corresponding device\rockchip\rkxxx directory, check if there is already a recovery.fstab_AB file. If it exists, skip this step directly. If there is no recovery.fstab_AB file, proceed with the following steps:

- Added recovery fstab file for AB: recovery.fstab_AB

The main difference from the corresponding fstab.rk30board lies in the addition of the slotselect mount parameter for AB partitions (system/vendor/odm/product partitions added with slotselect parameters), while simultaneously incorporating the partition nodes for external_sd, frp, parameter, baseparameter, and resource, and modifying the mount options for the data partition.

The following is a reference file:

device\rockchip\rk3399\rk3399_Android12\recovery.fstab_AB:



Import the A/B configuration into the BoardConfig under device\rockchip\rkxxx, including TARGET_RECOVERY_FSTAB, pointing it to the newly created recovery.fstab_AB file. A reference example is as follows:

```
     device\rockchip\rk3399\rk3399_Android12\BoardConfig.mk:
     diff --git a/ rk3399/rk3399_Android12/BoardConfig.mk b/
rk3399/rk3399_Android12/BoardConfig.mk
     index 1e78940..159a3b6 100755
     --- a/ rk3399/rk3399_Android12/BoardConfig.mk
     +++ b/ rk3399/rk3399_Android12/BoardConfig.mk
     +# AB image definition
     +BOARD_USES_AB_IMAGE := true
     +BOARD_ROCKCHIP_VIRTUAL_AB_ENABLE := false
     +
     +ifeq ($(strip $(BOARD_USES_AB_IMAGE)), true)
     +    include device/rockchip/common/BoardConfig_AB.mk
     +    TARGET_RECOVERY_FSTAB := device/rockchip/
rk3399/rk3399_Android11/recovery.fstab_AB
     +endif
```

2.1.2 Android 13 System Configuration

The configuration of Android 13 system includes:

1. In device\rockchip\rkxxx\BoardConfig.mk enable the AB update configuration item

Set the BOARD_USES_AB_IMAGE configuration to true, defaults to false. Taking rk3326 as an example:

```
   vim device/rockchip/rk3326_s/BoardConfig.mk
   #AB image definition
   -BOARD_USES_AB_IMAGE := false
   +BOARD_USES_AB_IMAGE := true
```

Note:

- If you want to enable the virtual AB feature (not recommended, may impact upgrade performance), you need to configure both BOARD_USES_AB_IMAGE and BOARD_ROCKCHIP_VIRTUAL_AB_ENABLE to true, i.e.:

```
-BOARD_USES_AB_IMAGE := false
-BOARD_ROCKCHIP_VIRTUAL_AB_ENABLE := false
+BOARD_USES_AB_IMAGE := true
+BOARD_ROCKCHIP_VIRTUAL_AB_ENABLE := true
```

- Starting from Android 13, support for the compressed virtual AB feature is available. To enable the compressed virtual AB feature (not recommended, as it may impact upgrade performance), all of BOARD_USES_AB_IMAGE, BOARD_ROCKCHIP_VIRTUAL_AB_ENABLE, and BOARD_ROCKCHIP_VIRTUAL_AB_COMPRESSION must be configured to true, i.e.:

```
-BOARD_USES_AB_IMAGE := false
-BOARD_ROCKCHIP_VIRTUAL_AB_ENABLE := false
+BOARD_USES_AB_IMAGE := true
+BOARD_ROCKCHIP_VIRTUAL_AB_ENABLE := true
+BOARD_ROCKCHIP_VIRTUAL_AB_COMPRESSION := true
```

2. In the corresponding device\rockchip\rkxxx directory, verify whether the recovery.fstab_AB file exists. If it already exists, skip this step directly. If the recovery.fstab_AB file does not exist, proceed with the following steps:

- Added recovery fstab file for AB: recovery.fstab_AB

The main difference from the corresponding fstab.rk30board lies in the addition of the slotselect mount parameter for AB partitions (system/vendor/odm/product partitions added with slotselect parameters), while incorporating partition nodes for exernal_sd, frp, parameter, baseparameter, and resource, and modifying the mount options for the data partition.

A reference file is as follows:

device\rockchip\rk3399\rk3399_Android12\recovery.fstab_AB:

```
# Android fstab file.
#<src>                                  <mnt_point>         <type>    <mnt_flags and options>                       <fs_mgr_flags>
# The filesystem that contains the filesystem checker binary (typically /system) cannot
# specify MF_CHECK, and must come before any filesystems that do specify MF_CHECK
system /system ext4 ro,barrier=1 wait,slotselect,logical,first_stage_mount
vendor /vendor ext4 ro,barrier=1 wait,slotselect,logical,first_stage_mount
odm    /odm    ext4 ro,barrier=1 wait,slotselect,logical,first_stage_mount
product    /product    ext4 ro,barrier=1 wait,slotselect,logical,first_stage_mount
system_ext   /system_ext    ext4 ro,barrier=1 wait,slotselect,logical,first_stage_mount
/dev/block/by-name/metadata /metadata ext4 nodev,noatime,nosuid,discard,sync wait,formattable,first_stage_mount
/dev/block/by-name/misc         /misc             emmc      defaults    defaults
/dev/block/by-name/cache         /cache             ext4      noatime,nodiratime,nosuid,nodev,noauto_da_alloc,discard               wait,check

/dev/block/mmcblk0p1                                    /mnt/external_sd     vfat            /dev/block/mmcblk0                 defaults
/dev/block/by-name/frp                  /frp                emmc            defaults                defaults
/dev/block/by-name/baseparamer          /baseparamer        emmc            defaults                defaults
/dev/block/by-name/backup               /backup             emmc            defaults                defaults

/dev/block/zram0                                none                swap      defaults                                                zramsize=50%

/dev/block/by-name/userdata             /data               f2fs            defaults                defaults
~
~
```

- Import the A/B configuration in BoardConfig under device\rockchip\rkxxx, including TARGET_RECOVERY_FSTAB, pointing to the newly created recovery.fstab_AB file. A reference example is as follows:

```
    device\rockchip\rk3399\rk3399_Android12\BoardConfig.mk:
    diff --git a/ rk3399/rk3399_Android12/BoardConfig.mk b/
rk3399/rk3399_Android12/BoardConfig.mk
    index 1e78940..159a3b6 100755
    --- a/ rk3399/rk3399_Android12/BoardConfig.mk
    +++ b/ rk3399/rk3399_Android12/BoardConfig.mk
    +# AB image definition
    +BOARD_USES_AB_IMAGE := true
    +BOARD_ROCKCHIP_VIRTUAL_AB_ENABLE := false
    +
    +ifeq ($(strip $(BOARD_USES_AB_IMAGE)), true)
    +    include device/rockchip/common/BoardConfig_AB.mk
    +    TARGET_RECOVERY_FSTAB := device/rockchip/
rk3399/rk3399_Android11/recovery.fstab_AB
    +endif
```

2.1.3 Android >=14 System Configuration

Android 14 system configuration includes:

1. Enable the corresponding A/B update configuration under device\rockchip\

Set the `BOARD_USES_AB_IMAGE` configuration to true, with the default being false. For example:

```
   diff --git a/rk3588_u/BoardConfig.mk b/rk3588_u/BoardConfig.mk
   index e30391c..c7a86ce 100644
   --- a/rk3588_u/BoardConfig.mk
   +++ b/rk3588_u/BoardConfig.mk
   @@ -14,7 +14,10 @@
   +
   +BOARD_USES_AB_IMAGE := true
```

Note:

- If you want to enable the virtual AB feature (not recommended, as it may affect upgrade performance), you need to configure both BOARD_USES_AB_IMAGE and BOARD_ROCKCHIP_VIRTUAL_AB_ENABLE to true, i.e.:

```
    +BOARD_USES_AB_IMAGE := true
    +BOARD_ROCKCHIP_VIRTUAL_AB_ENABLE := true
```

- If you need to enable the compressed virtual AB feature (targeted at GKI versions, not recommended, may impact upgrade performance), you need to configure BOARD_USES_AB_IMAGE, BOARD_ROCKCHIP_VIRTUAL_AB_ENABLE, and BOARD_ROCKCHIP_VIRTUAL_AB_COMPRESSION to true, i.e.:

```
    +BOARD_USES_AB_IMAGE := true
    +BOARD_ROCKCHIP_VIRTUAL_AB_ENABLE := true
    +BOARD_ROCKCHIP_VIRTUAL_AB_COMPRESSION := true
```

2.2 Kernel DTS Configuration

For Android >=12, the kernel requires no configuration.

2.3 U-Boot Configuration

In U-Boot, for the configuration files of specific chips, add the CONFIG_ANDROID_AB=y configuration item, as shown in the following screenshot:

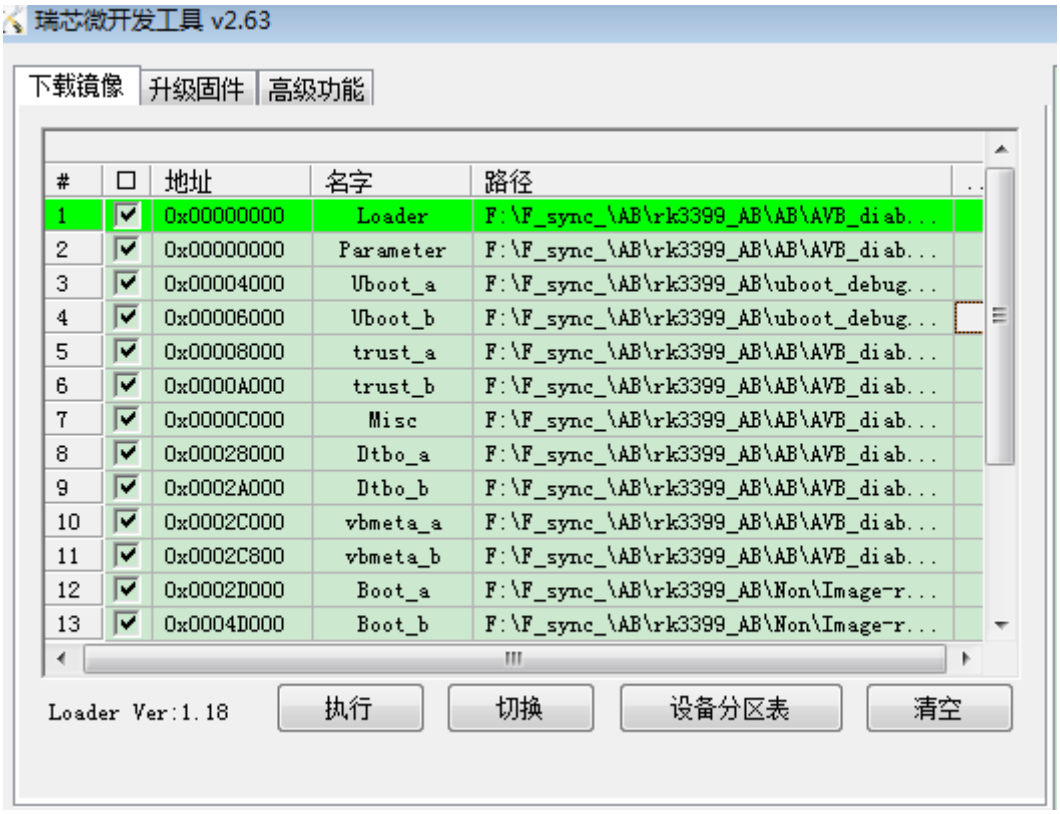
```
diff --git a/configs/rk3326_defconfig b/configs/rk3326_defconfig
old mode 100644
new mode 100755
index 0465f23..d64648c
--- a/configs/rk3326_defconfig
+++ b/configs/rk3326_defconfig
@@ -114,3 +114,4 @@ CONFIG_OPTEE_CLIENT=y
 CONFIG_OPTEE_V2=y
 CONFIG_OPTEE_ALWAYS_USE_SECURITY_PARTITION=y
 CONFIG_TEST_ROCKCHIP=y
+CONFIG_ANDROID_AB=y
```

2.4 Add Partitions via Advanced Flashing Tool

Download tool adds a download entry for B partition, while identical firmware is flashed to all AB partitions.

For Android 12 and above, after completing the compilation following the instructions in "3.AB System and OTA Package Compilation", the config.cfg file is automatically generated under the rockdev/xxxx/ directory. Replace the identically named file under the RKDevTool tool directory with this configuration file, then reopen RKDevTool. No manual configuration is required.

A screenshot of the flashing tool is shown below:



Note, for chips such as rk3566/rk3568/rk3588, if there is no trust partition, the trust_a and trust_b partitions are not required.

3. AB System and OTA Package Compilation

Compile AB System Firmware Steps (Must follow the steps below strictly):

Method 1:

```
source build/envsetup.sh
lunch XXX
make installclean -j16  OR make clean -j16
make -j16
make dist -j16
mkimage_ab.sh  ota
```

Method II:

```
source build/envsetup.sh
lunch XXX
./build.sh -ABUCKuop
```

Note:

1. After enabling AB, the first compilation requires executing 'make clean' or deleting the out directory before building. When switching between AB and non-AB configurations for compilation, a 'make clean' operation or out directory deletion must be performed prior to each build.
2. The compilation method for the difference package is consistent with that of non-AB.
3. Detailed compilation methods for full packages and difference packages can be found in 《Rockchip_Developer_Guide_Android_Recovery_CN》. The AB OTA update validation method should refer to the description in Section 4 "Validation Method (Client and Server)" of this document.
4. Rockchip has supported the GRF platform starting from Android 15 (i.e., >=Android 15). For the compilation methods of OTA full package and difference package for the Android GRF platform, please refer to 《Rockchip_Developer_Guide_Android_GRF_OTA》.

4. Validation Methods (Client and Server)

During the seamless upgrade process, the upgrade package can be downloaded and upgraded simultaneously. At this point, an HTTP server and an upgrade client are required.

Please use the following update_device.py for validation. Regarding the productized upgrade client and upgrade server, customers need to set them up independently. The upgrade client can reference the update_engine_client provided by default in Android. For the usage method of update_engine_client, please refer to the validation approach in update_device.py. This script invokes update_engine_client via adb to ultimately perform the upgrade.

Validation Method: update_device.py

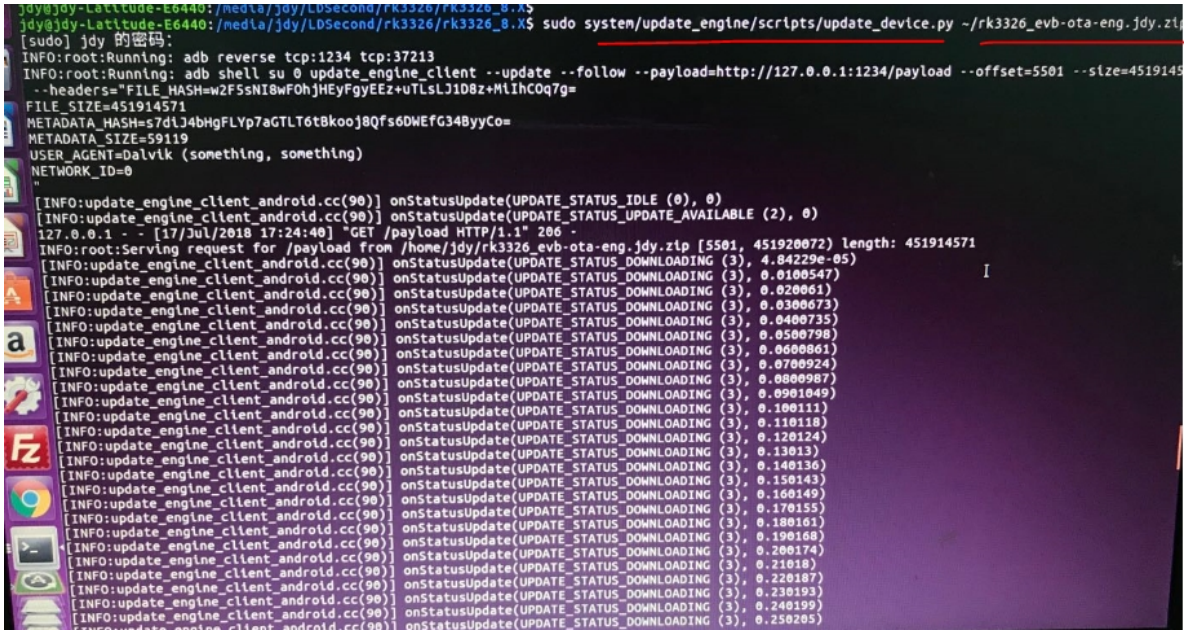
The update_device.py script uses ADB to turn the host into an HTTP server, then invokes update_engine_client to achieve seamless updates.

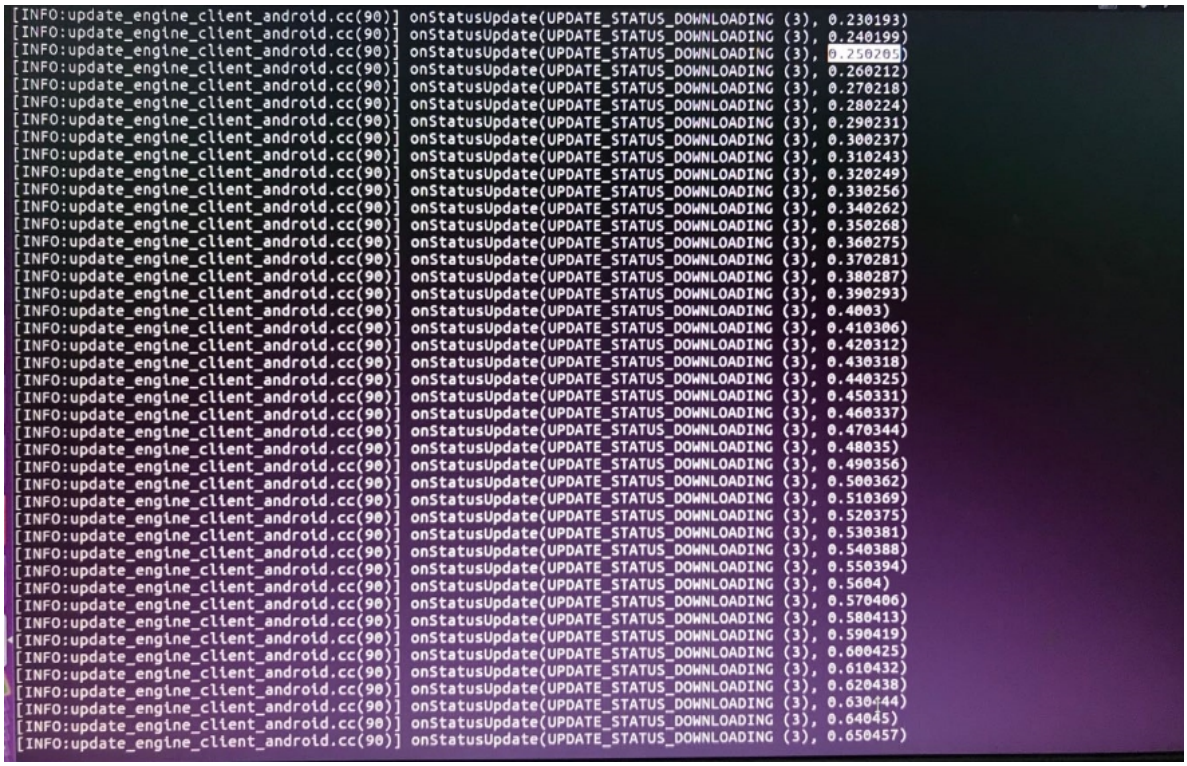
Usage method as follows:

Execute the following commands on the host (Ensure that the ADB functionality and Python environment are working properly):

```
system/update_engine/scripts/update_device.py {update package name}
```

Example as follows:









The above screenshot illustrates the complete upgrade process. After a successful upgrade, the system will print the information: UPDATE_STATUS_UPDATED_NEED_REBOOT. As shown in the above screenshot. Manually restarting the device will switch it to the newly upgraded system.

5. Notes

1. When packaging to generate update.img, ensure to modify the package-file to add the corresponding partitions.

For example, trust and uboot are configured as follows, other partitions follow a similar configuration.

```
trust_a       Image/trust.img
trust_b       Image/trust.img
uboot_a       Image/uboot.img
uboot_b       Image/uboot.img
```

For rk3566/rk3568/rk3588 and other chips, if there is no trust partition, then the trust_a and trust_b partitions are not required.

6. A/B miniloader OTA

Miniloader is the primary boot loader for the Rockchip platform. Under normal circumstances, this firmware does not require an OTA update, and it is not recommended to do so. In special cases, if an OTA update for the Miniloader is required, the solution introduced in this section can be implemented for Rockchip A/B system platforms.

It should be noted that the miniloader OTA update package has no relation to the normal A/B update package and is completely independent. This means that when an upgrade of the miniloader in the A/B system is required, the corresponding miniloader update package must be generated following the method introduced in Section 6.1. Additionally, the client must be developed according to the instructions in Section 6.2 (executing the reference commands from Section 6.2 during miniloader package upgrade will trigger entering recovery mode to upgrade the A/B system's miniloader, after which the system will automatically reboot).

6.1 rkloader Upgrade Package Build

For devices >= Android 12, after completing firmware compilation via OTA, the automatically generated update_loader.zip file under the rockdev directory is the rk loader OTA update package.

6.2 Upgrade Command Reference

Upgrading the client APK can be implemented by referring to the following command methods:

```
adb root
adb push update_loader.zip /cache/
adb shell "echo \"--fw_rkloader=/cache/update_loader.zip\" >
/cache/recovery/command"
adb reboot recovery
```